UAV MOTION PLANNING HANDOUT

Sampling-Based Motion Planning

Mathematical foundations, probability bounds, convergence, and trajectory limitations

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Central claim. Sampling-based planners avoid explicitly enumerating a full grid. Under robust-feasibility and sampling assumptions, their probability of finding a solution approaches one. This is an asymptotic statement: finite-sample failure remains possible, and dimension still enters through volume ratios, connection radii, and neighbor-search cost.

1 Configuration-space formulation

Let $\mathcal{C} \subset \mathbb{R}^d$ be the configuration space and $\mathcal{C}_{\text{obs}}$ the collision set. The free space is

$$\mathcal{C}_{\text{free}} = \text{cl}(\mathcal{C} \setminus \mathcal{C}_{\text{obs}}).$$

Given start $x_s \in \mathcal{C}_{\text{free}}$ and goal region $\mathcal{C}_g \subset \mathcal{C}_{\text{free}}$, a feasible path is a continuous map

$$\sigma : [0,1] \to \mathcal{C}_{\text{free}}, \qquad \sigma(0) = x_s, \quad \sigma(1) \in \mathcal{C}_g.$$

For geometric planning, a common objective is arc length

$$J(\sigma) = \int_0^1 \|\dot{\sigma}(t)\|_2 \, dt.$$

Sampling-based planners construct a finite graph or tree from collision-free samples and local connections, then return a piecewise path through that structure.

Definition 1 (Strong clearance). *A path σ has strong clearance $\delta > 0$ if*

$$B(\sigma(t), \delta) \subset \mathcal{C}_{\text{free}} \qquad \forall t \in [0,1].$$

This is the robust-feasibility assumption behind the elementary completeness proof. A zero-clearance passage may be feasible mathematically but have sampling probability zero.

2 Sampling model and positive-measure sets

Let samples $X_1, \dots, X_n$ be independent and identically distributed with density f on $\mathcal{C}_{\text{free}}$. For measurable $A \subset \mathcal{C}_{\text{free}}$,

$$\mathbb{P}(X_k \in A) = \int_A f(x) \, dx =: p_A.$$

Uniform sampling gives

$$p_A = \frac{\text{vol}(A)}{\text{vol}(\mathcal{C}_{\text{free}})}.$$

More generally, proofs require the sampling distribution to assign positive probability to every covering region needed by a robust solution. A sufficient condition is a density lower bound $f(x) \geq f_{\min} > 0$ on relevant free-space neighborhoods.

3 The probability of missing one region

Fix a measurable region A with $p_A > 0$. By independence,

$$\mathbb{P}(X_1 \notin A, \ldots, X_n \notin A) = (1 - p_A)^n.$$

Using $\log(1-p) \le -p$ for $p \in [0,1)$,

$$(1 - p_A)^n = \exp\big(n \log(1 - p_A)\big) \le e^{-np_A}.$$

Therefore

$$\mathbb{P}(A \text{ is hit by time } n) = 1 - (1 - p_A)^n \xrightarrow[n\to\infty]{} 1.$$

For every finite n, however, $(1 - p_A)^n > 0$ whenever $p_A < 1$. Thus "probabilistically complete" never means guaranteed success after a fixed number of samples.

To make the miss probability at most α, it is sufficient that

$$e^{-np_A} \le \alpha \quad \iff \quad n \ge \frac{1}{p_A} \log \frac{1}{\alpha}.$$

4 Covering a robust path

Assume σ has clearance δ. Choose balls

$$B_i = B(z_i, r), \qquad z_i = \sigma(t_i), \qquad r < \delta,$$

whose centers appear in path order and whose spacing is small enough that the local planner can connect a point in B_i to a point in B_{i+1} collision-free. Compactness of $\sigma([0,1])$ implies that a finite subcover $B_1, \ldots, B_m$ exists.

Let E_i be the event that B_i receives no sample. If $p_i = \mathbb{P}(X \in B_i) > 0$, then

$$\mathbb{P}(E_i) = (1 - p_i)^n \le e^{-np_i}.$$

Failure to represent the covered path implies that at least one required ball is empty. By the union bound,

$$\mathbb{P}(\text{failure by } n) \le \mathbb{P}\left(\bigcup_{i=1}^{m} E_i\right) \le \sum_{i=1}^{m} \mathbb{P}(E_i) \le \sum_{i=1}^{m} e^{-np_i}.$$

Let $p_{\min} = \min_i p_i > 0$. Then

$$\boxed{\mathbb{P}(\text{failure by } n) \le m e^{-np_{\min}} \to 0.}$$

Consequently, a sufficient finite-sample condition for failure probability at most α is

$$\boxed{n \ge \frac{1}{p_{\min}} \log \frac{m}{\alpha}.}$$

Theorem 1 (Probabilistic completeness, proof template)**.** *Suppose a robust feasible path admits a finite ordered cover by positive-probability regions, the sampler is independent with nonzero probability on each region, and the local connection rule eventually links samples in consecutive regions. Then the probability that the planner has found a feasible path approaches one as $n \to \infty$.*

Proof. The probability that any covering region remains empty is bounded by $me^{-np_{\min}}$, which approaches zero. On the complementary event every region is occupied. By the local-connection assumption, the ordered samples form a collision-free graph path from start to goal. Hence the planner's failure probability also approaches zero. □

5 A numerical finite-sample example

Suppose a robust route is covered by $m = 20$ regions and the smallest region has sampling probability $p_{\min} = 0.002$. To make the union-bound failure estimate at most $\alpha = 0.01$,

$$n \geq \frac{1}{0.002} \log \frac{20}{0.01} = 500 \log(2000) \approx 3801.$$

This is sufficient, not necessary; the union bound may be conservative. It nonetheless exposes the correct dependencies: rarer required regions, more covering regions, or stronger confidence all increase the sample requirement.

6 Where dimension re-enters

For a d-dimensional Euclidean ball of radius r,

$$\mathrm{vol}(B_d(r)) = \zeta_d r^d, \qquad \zeta_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$$

Under uniform sampling in free volume V_f,

$$p_i \approx \frac{\zeta_d r^d}{V_f}.$$

Substituting into the finite-sample bound gives the scaling

$$n \gtrsim \frac{V_f}{\zeta_d r^d} \log \frac{m}{\alpha}.$$

For fixed $r < 1$ in normalized coordinates, r^{-d} grows exponentially. Sampling avoids storing every cell of a predetermined q^d grid, but it does not abolish the curse of dimensionality: useful neighborhoods occupy rapidly shrinking volume fractions.

If dimensions have unequal characteristic widths or clearances, normalizing coordinates matters. A Euclidean ball in poorly scaled coordinates may allocate samples inefficiently or define meaningless nearest neighbors.

7 PRM and RRT: what the proof additionally needs

Probabilistic Roadmap (PRM)

PRM samples free configurations, connects nearby pairs when a local planner succeeds, and searches the resulting graph. Completeness requires both occupancy of covering regions and a connection rule that produces the needed edges. A fixed connection radius can work for feasibility under suitable clearance, but asymptotically optimal variants use a radius that changes with n.

Rapidly-exploring Random Tree (RRT)

RRT repeatedly samples x_{rand}, finds a nearest tree vertex, steers toward the sample, and adds the collision-free extension. Its completeness proof is not merely the independent occupancy argument because tree extensions depend on previous samples. The key is a uniformly positive probability of advancing from one attraction region to the next along a robust path. Repeated trials then drive the probability of never advancing to zero.

Algorithm 1 Generic sampling-based tree expansion

$T \leftarrow \{x_s\}$
for $k = 1, \dots, n$ **do**
  $x_{\text{rand}} \sim \mu$ on $\mathcal{C}_{\text{free}}$
  $x_{\text{near}} \leftarrow \text{Nearest}(T, x_{\text{rand}})$
  $x_{\text{new}} \leftarrow \text{Steer}(x_{\text{near}}, x_{\text{rand}}, \eta)$
  if $\text{CollisionFree}(x_{\text{near}}, x_{\text{new}})$ **then**
    add x_{new} and edge $(x_{\text{near}}, x_{\text{new}})$ to T

Biased sampling, goal bias, informed subsets, and learned samplers can accelerate planning, but completeness is retained only if the sampling scheme preserves sufficient probability of visiting every region required by some robust solution.

8 Completeness is not optimality

Let J_n be the cost of the best solution in the structure after n samples and J^* the continuous optimum.

Definition 2 (Asymptotic optimality). *A planner is asymptotically optimal if*

$$\mathbb{P}\left(\lim_{n\to\infty} J_n = J^*\right) = 1.$$

Ordinary RRT is probabilistically complete but generally not asymptotically optimal: early branches can lock in poor ancestry. RRT* adds rewiring; PRM* uses a sufficiently rich random geometric graph.

For n samples in d dimensions, a standard asymptotic connection scale is

$$r_n = \min\left\{\eta,\ \gamma_{\text{conn}}\left(\frac{\log n}{n}\right)^{1/d}\right\},$$

with γ_{conn} above a problem-dependent threshold. The radius shrinks so edges become local, but slowly enough that the graph remains sufficiently connected. A k-nearest alternative uses

$$k_n \geq k_0 \log n.$$

The constants and theorem assumptions depend on the algorithm, free-space regularity, cost functional, steering method, and boundary behavior. Writing only $r_n \propto (\log n/n)^{1/d}$ without these assumptions is incomplete.

9 Why the returned path is jagged

A graph path through vertices $x_0, \dots, x_N$ is commonly interpolated segment by segment:

$$\gamma_i(s) = (1-s)x_i + s x_{i+1}, \qquad s \in [0,1].$$

Within segment i,

$$\frac{d\gamma_i}{ds} = x_{i+1} - x_i, \qquad \frac{d^2\gamma_i}{ds^2} = 0.$$

At waypoint x_i, the one-sided derivatives are

$$\dot\gamma(t_i^-) = \frac{x_i - x_{i-1}}{\Delta t_{i-1}}, \qquad \dot\gamma(t_i^+) = \frac{x_{i+1} - x_i}{\Delta t_i}.$$

They are unequal in general. Velocity therefore jumps, and acceleration contains an impulse in the distributional sense:

$$\ddot{\gamma} \supset \big(\dot{\gamma}(t_i^+) - \dot{\gamma}(t_i^-)\big)\delta(t - t_i).$$

An aircraft with bounded thrust and acceleration cannot track this command exactly. Geometric feasibility is not dynamic feasibility.

10 Smoothing and its mathematical obligations

Shortcutting or spline fitting can reduce length and derivative discontinuities, but an unconstrained smoother may cut through obstacles. A safe post-processing problem has the form

$$\min_p \int_0^T \|p^{(k)}(t)\|^2 dt$$

subject to

$$p(t) \in \mathcal{C}_{\text{free}}, \quad p(0) = x_s, \quad p(T) \in \mathcal{C}_g, \quad \|\dot{p}\| \leq v_{\max}, \quad \|\ddot{p}\| \leq a_{\max}.$$

Using a fixed convex safe corridor can turn polynomial smoothing into a convex quadratic program. Without a safe corridor or exact collision constraints, smoothing must be collision-checked continuously or conservatively certified.

Time parameterization is separate from geometric smoothing. Even a C^2 geometric curve may violate velocity or acceleration limits if traversed too quickly.

11 What can and cannot be claimed

Claim	Precise interpretation
"Scales to high dimensions"	Avoids enumerating a full grid and often works better empirically; sample complexity can still degrade exponentially through volume ratios.
Probabilistically complete	Success probability tends to one under robust feasibility, sampling support, and connection assumptions.
Finite-sample guarantee	Requires an explicit bound such as $me^{-np_{\min}}$; completeness alone gives no fixed-n certainty.
Asymptotically optimal	Best-path cost converges almost surely to J^* under stronger radius, rewiring, regularity, and cost assumptions.
Collision-free path	Edges passed the chosen collision checker; this does not automatically imply dynamic feasibility or tracking robustness.
Jagged path	Piecewise interpolation generally has derivative jumps and needs smoothing plus time parameterization.

12 Summary

The elementary probability chain is

$$\boxed{p_i > 0 \Longrightarrow \mathbb{P}(B_i \text{ missed}) = (1 - p_i)^n \leq e^{-np_i} \Longrightarrow \mathbb{P}(\text{failure}) \leq \sum_i e^{-np_i} \to 0.}$$

Its force comes from explicit assumptions: a robust path, finitely many positive-measure covering regions, sampling support, and successful local connections. Its limitation is equally explicit: every finite sample budget retains nonzero failure probability, and high dimension shrinks useful volume. Sampling-based planning exchanges exhaustive discretization for randomized exploration; it does not exchange mathematics for luck.